

Youssef Naciri Zerta

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Fes (Morocco) until February 2027 | Cambridge (UK) February–August 2027

RESEARCH FOCUS

My aim is to move a cell from its current state to a chosen target state with an intervention *designed from a model* rather than found by screening. That needs two things: a model that predicts the effect of interventions nobody has tried yet, and a clear account of which experiments make such a model reliable enough to act on. My thesis works on both. It asks which kinds of intervention can identify regulatory mechanism at all, in simulated gene networks where the truth is known, and whether a causal model of regulatory state learned from perturbation data in differentiating stem cells can predict interventions it has not seen. Alongside the thesis, I lead the development of AnnNet, a data structure for biological networks whose interactions change with context. I also audit claims drawn from learned representations: in one audit, a standard interpretability test reported discoveries that turned out to be an artifact of how its null distribution was thresholded.

RESEARCH

Research Thesis: From Perturbation Data to Reliable Cell-State Control

2026–2027

EMBL-EBI, Saez Lab, with the Wellcome Sanger Institute — with Rory Maizels – *research project 09/2026–02/2027 (remote, part time); thesis 02–08/2027 (full time, on site)*

- **Which interventions can identify regulatory mechanism?** I simulate published gene regulatory networks with ordinary differential equations and realistic count noise, so that the true mechanism is known. I then ask which intervention protocols let competing mechanisms be told apart, and whether the ambiguity that remains would change a prediction or an intervention decision. The current test compares gene knockdowns and knockouts with calibrated, state-dependent additions of production, similar to what an enhancer-driven transcription-factor construct delivers. Preprint planned for November 2026.
- **Can a causal model of regulatory state predict new interventions?** Using perturbation data from differentiating stem cells, I will learn a causal representation of regulatory state and test it where it matters for control: predicting interventions held out from training, and choosing interventions that reach a target cell state.

AnnNet and SysBioVerse — Lead Developer

2025–Present

EMBL-EBI, Saez Lab, with Open Targets – *co-first-author preprint forthcoming (target: Bioinformatics)*

- **AnnNet** is AnnData for biological networks: one annotated container for directed, signed, hyperedge, multilayer and condition-specific structure, with lossless storage and interoperability with NetworkX, igraph, graph-tool and PyTorch Geometric. I designed and built the core: the incidence representation, the annotation model, the storage format and the library interfaces.
- Used by groups at EMBL-EBI and Imperial College London, and developed with scverse core developers so that it fits the single-cell Python ecosystem.
- Now extending into SysBioVerse (2026–), a network-biology consortium with scverse: one interface for regulatory, signalling, metabolic, cell–cell and multi-omic causal inference, so that methods which model biological structure differently can be compared and combined instead of being reimplemented for each paper.

Sparsity, Not Alignment: How a Standard Interpretability Test Manufactures Discovery

2026

Sole author – *accepted, NeurIPS 2026 workshop; [paper](#)*

- Tested whether sparse-autoencoder features of a genomic foundation model correspond to regulatory mechanism, using saturation-mutagenesis MPRA (massively parallel reporter assay) data as per-base ground truth. Run the standard way, the test reports a discovery that passes every control the literature uses: label shuffling, repeat masking, a likelihood baseline, three seeds, two autoencoder families and two architectures.
- The discovery is an artifact. The test thresholds one tail of a null distribution, and the same FDR (false discovery rate) procedure applied to the other tail finds more “discoveries”. The two distributions differ in spread, not location, and the effect grows with dictionary size, because sparser units have noisier scores.
- Proposed four diagnostics that cost nothing extra, a studentised repair, and a noise ceiling showing that the accompanying reproducibility failure comes from the size of the ground-truth dataset, not from the decomposition.

Multiplex Bacterial Interaction Networks

2024–Present

Université Euromed de Fès – *experimental systems biology; first author; [preprint](#)*

, under review at *ISME Communications*

- Designed and ran a trait-resolved interaction study across 60 soil bacterial strains: pairwise assays producing 12 directed interaction layers, analysed as one multilayer network rather than a single graph.

- Showed that the sign and strength of an interaction depend on context rather than being fixed for a pair of strains, and placed the strains in a low-dimensional space of antagonistic, metabolic and morphogenetic strategies. This representational problem is what led to AnnNet.

SampleCLR — Patient-Level Representations from Single-Cell Transcriptomics

2025

Helmholtz Munich, Lücken Lab – *contrastive and attention-based sample representation*; [preprint](#)

- Contributed the interpretability and transfer-learning analyses and parts of the architecture design for a model that learns patient-level representations from single-cell data; the paper reports state-of-the-art disease-stratification performance.
- Tested whether the representations transfer across biological systems rather than only fitting the training benchmark.

Also: analysis pipeline for detecting biomarker combinations from epigenomic data, MD Anderson Cancer Center / KAUST, 2024–2025.

PUBLICATIONS & PREPRINTS

- **Y. Zerta**, et al. *Multilayer interaction networks reveal multidimensional ecological strategies and context-dependent competition in soil actinobacteria* First author. bioRxiv (2026), doi:10.64898/2026.09.16.752070. Under review at *ISME Communications*.
- **Y. Naciri Zerta**. *Sparsity, not alignment: how a standard interpretability test manufactures discovery in a genomic foundation model*. Sole author. accepted, NeurIPS 2026 workshop. [paper](#)
- *AnnNet: a unified annotated-graph structure for biological networks*. Co-first author. Preprint forthcoming (target: *Bioinformatics*).
- *Great patients embed alike: contrastive learning for sample representation from single-cell data*. Middle author. [\[preprint\]](#)

EDUCATION

Ingénieur d'État en Biotechnologies (State Engineer, Biotechnology)

Expected 2027

Université Euromed de Fès (UEMF), Morocco – *five-year engineering degree; Master's-equivalent, 300 ECTS (European Credit Transfer and Accumulation System)*

Relevant coursework: Systems Biology, Synthetic Microbiology, Recombinant DNA Technology, Statistical Genetics, Analytic Chemistry, Organic Chemistry, Probability & Statistics.

METHODS

Causal inference & representation learning: identifiability, interventional inference, causal representation learning, sparse autoencoders and other latent-variable models, contrastive learning, mechanistic interpretability.

Modelling & statistics: ODE (ordinary differential equation) models of gene regulation, Bayesian model fitting, high-dimensional and sparse inference.

Evaluation design: interventional ground truth, null calibration, permutation inference, multiple testing, group-structured cross-validation, leakage diagnosis, noise ceilings, planted nulls.

Genomics & regulation: gene regulatory networks, perturbational genomics, single-cell transcriptomics, MPRA data, statistical genetics.

Systems biology: mechanistic and dynamical modelling, multilayer and hypergraph formalisms, prior-knowledge integration.

Engineering: Python (PyTorch, NumPy, scikit-learn), scientific library design, open-source maintenance, GPU training, cloud computing, Docker.

Wet lab: molecular cloning, recombinant DNA, genome engineering, high-throughput microbial culture and interaction assays, analytical chemistry.

Languages: English & French (fluent), Arabic & Tamazight (native), Spanish (basic).